

**BLOOD TRIBE AGDM, AB, Canada**

WMO: 715170

Lat: **49.5588N**Lon: **113.0633W**Elev: **999**StdP: **89.89**Time Zone: **-7.00 (NAM)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -28.4      | -24.9 | -31.7                         | 0.2 | -28.3 | -27.9 | 0.3 | -24.9 | 19.4                  | 2.0  | 18.4 | 2.7  | 3.4                   | 230  | 0.748 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.8                   | 30.7            | 16.2 | 28.8 | 15.8 | 26.9 | 15.4 | 18.8                | 25.9 | 17.6 | 25.1 | 16.5 | 24.3 | 4.6                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.2                            | 13.0 | 22.3 | 14.8 | 11.9 | 21.0 | 13.6 | 11.0 | 19.3 | 57.8          | 26.1 | 53.7 | 25.1 | 50.0 | 24.4 | 25.6              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| (4) 16.4          | 14.5 | 12.8 | DB                         | -33.8 | 34.2               | 3.7 | 2.3                                                | -36.5 | 35.8       | -38.7 | 37.1       | -40.8 | 38.4       | -43.5 | 40.0 |  |
| (5)               |      |      | WB                         | -33.9 | 21.4               | 3.7 | 1.6                                                | -36.6 | 22.6       | -38.7 | 23.5       | -40.8 | 24.4       | -43.5 | 25.6 |  |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 5.4    | -4.7  | -6.1  | -0.4  | 4.9   | 10.4  | 14.3  | 17.7  | 16.7  | 12.3  | 5.8   | -0.4  | -6.5  |
|                                                                        | DBStd       | 10.50  | 8.60  | 8.70  | 7.32  | 4.88  | 4.31  | 3.14  | 2.96  | 3.34  | 4.57  | 5.41  | 7.34  | 8.60  |
|                                                                        | HDD10.0     | 2438   | 456   | 451   | 322   | 161   | 47    | 3     | 0     | 1     | 28    | 147   | 312   | 511   |
|                                                                        | HDD18.3     | 4772   | 714   | 683   | 579   | 402   | 246   | 127   | 47    | 71    | 185   | 387   | 561   | 769   |
|                                                                        | CDD10.0     | 766    | 1     | 0     | 1     | 8     | 61    | 131   | 239   | 209   | 97    | 17    | 2     | 0     |
|                                                                        | CDD18.3     | 58     | 0     | 0     | 0     | 0     | 1     | 5     | 28    | 21    | 4     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 1493   | 0     | 0     | 0     | 7     | 72    | 136   | 582   | 513   | 179   | 4     | 0     | 0     |
|                                                                        | CDH26.7     | 452    | 0     | 0     | 0     | 1     | 11    | 26    | 192   | 174   | 48    | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 5.3    | 6.6   | 5.5   | 5.6   | 5.6   | 5.0   | 5.0   | 4.0   | 3.9   | 4.6   | 5.5   | 6.2   | 6.2   |
| (15) Precipitation                                                     | PrecAvg     | 436    | 17    | 13    | 22    | 36    | 65    | 97    | 42    | 39    | 38    | 21    | 17    | 19    |
|                                                                        | PrecMax     | 687    | 31    | 41    | 52    | 82    | 125   | 307   | 151   | 96    | 132   | 74    | 37    | 39    |
|                                                                        | PrecMin     | 207    | 7     | 1     | 2     | 9     | 12    | 18    | 1     | 0     | 4     | 2     | 3     | 2     |
|                                                                        | PrecStd     | 114    | 6     | 10    | 12    | 17    | 30    | 64    | 34    | 24    | 29    | 16    | 9     | 8     |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 12.8  | 13.2  | 17.1  | 24.1  | 28.3  | 30.2  | 33.6  | 33.2  | 30.7  | 23.3  | 17.8  | 11.4  |
|                                                                        |             | MCWB   | 6.1   | 5.5   | 7.5   | 11.0  | 13.3  | 17.6  | 17.0  | 15.7  | 14.6  | 11.5  | 8.7   | 4.8   |
|                                                                        | 2%          | DB     | 9.8   | 10.1  | 14.2  | 19.6  | 25.4  | 26.7  | 30.8  | 30.7  | 28.0  | 20.1  | 13.9  | 8.7   |
|                                                                        |             | MCWB   | 4.5   | 4.0   | 6.2   | 9.0   | 12.6  | 15.7  | 17.0  | 15.6  | 13.9  | 10.3  | 7.1   | 3.8   |
|                                                                        | 5%          | DB     | 7.3   | 7.7   | 12.1  | 16.9  | 22.8  | 24.7  | 28.8  | 28.5  | 25.3  | 17.1  | 11.2  | 6.5   |
|                                                                        |             | MCWB   | 2.9   | 2.6   | 5.4   | 7.5   | 11.6  | 14.8  | 16.9  | 15.3  | 13.2  | 9.1   | 5.6   | 2.5   |
|                                                                        | 10%         | DB     | 5.5   | 5.5   | 9.7   | 14.0  | 20.1  | 22.5  | 26.7  | 26.3  | 22.2  | 14.3  | 9.0   | 4.7   |
|                                                                        |             | MCWB   | 1.9   | 1.4   | 4.0   | 6.4   | 10.4  | 13.7  | 16.6  | 15.0  | 12.2  | 7.8   | 4.2   | 1.2   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 6.8   | 6.2   | 8.5   | 11.2  | 14.4  | 19.5  | 21.4  | 19.0  | 16.2  | 12.7  | 9.8   | 6.1   |
|                                                                        |             | MCDB   | 11.7  | 11.9  | 15.4  | 22.4  | 25.1  | 26.5  | 27.9  | 26.2  | 25.6  | 21.3  | 16.6  | 10.4  |
|                                                                        | 2%          | WB     | 4.6   | 4.3   | 7.0   | 9.4   | 13.2  | 17.3  | 19.5  | 17.7  | 14.7  | 11.1  | 7.6   | 4.1   |
|                                                                        |             | MCDB   | 9.1   | 9.6   | 13.4  | 18.5  | 23.8  | 24.8  | 26.5  | 25.8  | 25.1  | 18.8  | 13.4  | 7.8   |
|                                                                        | 5%          | WB     | 3.2   | 2.8   | 5.6   | 8.0   | 12.1  | 15.6  | 18.3  | 16.6  | 13.7  | 9.6   | 5.8   | 2.6   |
|                                                                        |             | MCDB   | 6.9   | 7.2   | 11.4  | 16.3  | 21.7  | 22.6  | 25.4  | 25.2  | 23.8  | 16.2  | 10.7  | 6.2   |
|                                                                        | 10%         | WB     | 2.1   | 1.5   | 4.2   | 6.6   | 11.0  | 14.4  | 17.3  | 15.6  | 12.5  | 8.1   | 4.4   | 1.3   |
|                                                                        |             | MCDB   | 5.3   | 5.2   | 9.4   | 13.4  | 18.9  | 21.0  | 24.5  | 24.3  | 21.7  | 13.9  | 8.7   | 4.7   |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.4  | 11.6  | 11.9  | 13.0  | 13.5  | 13.1  | 15.8  | 16.2  | 14.7  | 12.7  | 11.0  | 10.5  |
|                                                                        | 5% DB       | MCDBR  | 11.0  | 13.3  | 15.4  | 18.5  | 18.5  | 17.5  | 19.9  | 21.1  | 20.9  | 16.8  | 13.8  | 10.8  |
|                                                                        |             | MCWBR  | 7.1   | 8.4   | 8.9   | 9.2   | 8.1   | 8.4   | 9.1   | 8.7   | 8.8   | 8.6   | 8.1   | 7.4   |
|                                                                        | 5% WB       | MCDBR  | 10.9  | 12.7  | 14.6  | 17.2  | 17.0  | 15.7  | 17.2  | 18.0  | 19.1  | 16.0  | 13.3  | 10.9  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.248 | 0.258 | 0.280 | 0.319 | 0.331 | 0.327 | 0.332 | 0.348 | 0.311 | 0.280 | 0.256 | 0.232 |
|                                                                        | taud        |        | 2.440 | 2.467 | 2.484 | 2.386 | 2.411 | 2.466 | 2.462 | 2.395 | 2.472 | 2.538 | 2.500 | 2.446 |
|                                                                        | Ebn at Noon |        | 834   | 912   | 943   | 929   | 928   | 931   | 921   | 888   | 893   | 873   | 819   | 810   |
|                                                                        | Edh at Noon |        | 59    | 74    | 88    | 108   | 111   | 106   | 105   | 107   | 89    | 69    | 56    | 51    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.24  | 2.25  | 3.50  | 4.84  | 5.77  | 6.43  | 7.08  | 5.62  | 4.00  | 2.49  | 1.40  | 0.99  |
|                                                                        | RadStd      |        | 0.10  | 0.17  | 0.29  | 0.33  | 0.49  | 0.47  | 0.37  | 0.34  | 0.33  | 0.25  | 0.11  | 0.10  |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (64 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page